Module 08: Planning in Robotics

Learning Objectives

1. Define **Planning** within the context of Robotics.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Robotics, we see Planning manifest in: * **Specific Example 1**: A Model Predictive Control (MPC) planner for an autonomous vehicle evaluates candidate steering and acceleration sequences over a 3-second receding horizon by simulating the vehicle's dynamics model forward in time and scoring each trajectory against a cost function encoding lane-center tracking, speed maintenance, and collision avoidance; this is precisely Active Inference planning through expected free energy evaluation -- each candidate policy's predicted sensory consequences (future positions) are compared against prior preferences (stay in lane, maintain speed), and the policy with minimum expected free energy is selected. * **Specific Example 2**: A trajectory optimization planner using iterative LQR (iLQR) computes locally optimal manipulation trajectories for a 7-DOF arm by alternating between a forward simulation pass (predicting the trajectory under current controls) and a backward pass (computing corrective feedback gains via dynamic programming); each iteration reduces the expected cost, which maps to expected free energy minimization -- the planner converges when prediction errors between the desired goal state and the simulated final state fall below the task-required tolerance.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.